MIL-C-81788(AS)
6 August 1970
Superseding AS-1072
31 May 1968

MILITARY SPECIFICATION

CARTRIDGES, DELAY, MARK 4 MOD 2, MARK 5 MOD 2 AND MARK 6 MOD 2

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This document governs the manufacture, assembly and preparation for delivery of the Cartridges, Delay, MARK 4 MOD 2, MARK 5 MOD 2 and MARK 6 MOD 2 and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this document to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-P-20444	Primer, Percussion, M42, Loading, Assembling and Packing
MIL-M-21383	Manganese Delay Composition
MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
MIL-P-22264	Powder, Ignition, Gasless

FSC 1377

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:10Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Naval Air Systems Command (CODE IDENT 30003)

LD 537902	Container, Inner, for Cartridges, Power Source (Formerly Code Ident 10001)
LD 537903	Container, Outer, for Cartridges, Power Source (Formerly Code Ident 10001)
DL 2519703	Cartridge, Delay, MARK 4 MOD 2
DL 2519704	Cartridge, Delay, MARK 5 MOD 2
DL 2519705	Cartridge, Delay, MARK 6 MOD 2
2518447	Holder and Primer Assembly
2518448	Delay, Loaded Assembly
2519703	Cartridge, Delay, MARK 4 MOD 2 Assembly
2519704	Cartridge, Delay, MARK 5 MOD 2 Assembly
2519705	Cartridge, Delay, MARK 6 MOD 2 Assembly
2838183	Test Set for Cartridges, Delay (.38 Caliber Size)
2838190	Sleeve

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:10Z
Check the source to verify that this is the current version before use.

Ordnance Corps of the Army (CODE IDENT 19203)

8799925	Primer, Percussion, M42, Assembly

PUBLICATIONS

Bureau of Naval Weapons (CODE IDENT 10001)

OP 5	Ammunition Ashore (This publication is not furnished to bidders and contractors by the procuring activity. Contractors should contact the Government Inspector for use of the publication.)
WR-43	Preparation of Quality Assurance Provisions

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-178	Transportation, Explosives and Other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should request the latest edition and supplements thereto.)

3. REQUIREMENTS

3.1 General. The three delay cartridges covered by this specification are identical in design except for delay time as specified in 3.9.1.

3.1.1 The MARK 4 MOD 2, MARK 5 MOD 2 and MARK 6 MOD 2 delay cartridges covered by this specification shall be manufactured in accordance with the drawings listed on DL 2519703, DL 2519704, and DL 2519705, respectively.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:10Z
Check the source to verify that this is the current version before use.

3.2 Materials. All materials used in the manufacture of the cartridges shall conform strictly with the specifications referenced on the respective Naval Air Systems Command drawings unless specific approval in writing covering a departure therefrom has been obtained from the Navy cognizant field activity for cartridges prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2.1 Preproduction sample. Unless otherwise specified by the procuring activity, a preproduction sample of 94 cartridges shall be manufactured using the same methods and procedures proposed for the production lot. The sample shall be tested as specified in 4.3.2 and is for the purpose of determining that the contractor, prior to starting production, is capable of producing items which comply with the technical requirements of the contract.

3.3 Primary components. For the purposes of this specification, the Primer, M42, MIL-P-20444; the Manganese Delay Composition, MIL-M-21383; and the Powder, Gasless Ignition, MIL-P-22264; are defined as primary components. See 4.3.3.

3.4 Primer. The primers shall be furnished from lots manufactured within the preceding 12 months. The primers shall be manufactured in accordance with Drawing 8799925, loaded with primer mixture as specified on Drawing 2518447, and accepted in accordance with MIL-P-20444. See 4.7.1.

3.4.1 X-ray. Primers shall be free of the following defects when examined by X-ray in accordance with 4.7.2: missing or abnormally thin charge; charge broken, cracked or porous; cracked or off-center anvil, or other misplaced or deformed inert parts; anvil abnormally close to primer cup dome; and all other observable defects in assembly.

3.5 Manganese delay composition. The manganese delay composition shall be manufactured in accordance with MIL-M-21383 and loaded in accordance with Drawing 2518448.

3.6 Gasless ignition powder. The gasless ignition powder A-1A shall be manufactured in accordance with MIL-P-22264, Class A and loaded in accordance with Drawing 2518448.

3.7 Propellant charge.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:10Z
Check the source to verify that this is the current version before use.

3.7.1 Quantity. All propellant charges for the MARK 4 MOD 2, MARK 5 MOD 2 and MARK 6 MOD 2 delay cartridges shall be as specified on Drawings 2519703, 2519704 and 2519705, respectively. Charges shall be determined by weight rather than by volume measurements.

3.7.2 Smokeless powder, handling and loading. Do not subject the smokeless powder to unnecessary handling or exposure to the atmosphere. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity to exceed 75 percent.

3.7.3 Smokeless powder, condition. The smokeless powder utilized shall exhibit no evidence of decomposition. See 4.7.3.

3.8 Cartridge inspection.

3.8.1 Visual inspection. Cartridges shall be free of the following visible defects: burrs, dents, deep scratches, split or cracked edges, damage to closure cup, sharp edges, defective sealant application around the crimped areas; and all other defects which could conceivably prevent entry into the firing chamber or adversely affect ballistic performance. See 4.7.4.

3.8.2 Leakage. Cartridge leakage rate shall not exceed 10^{-5} cc of air or gas per second when tested in accordance with 4.7.5.

3.8.3 X-ray. Cartridges shall be free of imperfections in assembly when examined by X-ray in accordance with 4.7.6.

3.8.3.1 Serial number identification. Cartridges shall be legibly identified by serial numbers in accordance with 4.7.6.

3.9 Ballistic requirements. Cartridges from each production lot, selected in accordance with 4.7.7, shall meet the requirements listed below when subjected to the test prescribed in 4.7.7.

3.9.1 Delay times. The delay cartridge shall function and produce the delay times within the limits shown below. Delay time is defined as the time from actuation of the firing mechanism to the start of pressure rise.

Delay Cartridge	Upper Limit (seconds)	Lower Limit (seconds)
MARK 4 MOD 2	0.95	0.61
MARK 5 MOD 2	2.53	1.62
MARK 6 MOD 2	3.80	2.43

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:10Z
Check the source to verify that this is the current version before use.

3.9.2 Minimum pressure. The cartridge shall deliver a peak pressure of not less than 1900 pounds per square inch.

3.9.3 Time to peak pressure. The time from start of pressure rise to peak pressure shall not exceed 0.020 second.

3.9.4 Misfire. There shall be no misfires, see 4.7.7.2.1.

3.10 Packaging, packing and marking. The packaging and packing of cartridges and the container markings shall be in accordance with Section 5. See 4.6.1.

3.11 Workmanship. Cartridges shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention should be directed to dimensions, finishes and coatings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions of MIL-STD-105.

4.3 Lots.

4.3.1 Inspection lots. Inspection lot definition, formation, and size shall be in accordance with MIL-STD-105.

4.3.2 Preproduction sample. Before entering into quantity production, an acceptable preproduction sample of 94 MARK 4 MOD 2, MARK 5 MOD 2 or MARK 6 MOD 2 delay cartridges shall be prepared. Of these cartridges, 84 shall be expended in the tests listed below and 10 shall be retained for investigative purposes. A preproduction sample acceptable for environmental and functional testing shall be defined as a sample which has met the requirements of paragraphs 3.3 through 3.8.3 of the specification. Preproduction sample acceptance shall consist of the following tests as prescribed in MIL-D-21625: six-foot drop (6); 28-day temperature

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:10Z
Check the source to verify that this is the current version before use.

and humidity cycling (18); vibration (12); and 15 "g" shock (12). These cartridges, after exposure to environmental treatments, shall be conditioned for a minimum of six hours at 70° ± 5° F together with 12 untreated cartridges and fired in accordance with 4.7.7.1. In addition, 12 cartridges shall be conditioned at -65°F and 12 shall be conditioned at +160°F. Temperature conditioning and firing of these cartridges shall be as specified in 4.7.7.1 for Group 1 and Group 2 samples, respectively. All cartridges shall fire and produce results within the limits set forth in paragraphs 3.9.1, 3.9.2 and 3.9.3. Any quantity production prior to notification by the contracting agency of preproduction sample acceptability shall be at the contractor's risk.

4.3.3 Production lot. Unless otherwise specified in the contract, the minimum and maximum production lot size shall be 501 and 3200 cartridges, respectively. Only primary components from a single lot shall be used in a production lot of cartridges; however, one primary component production lot may be used in more than one cartridge production lot.

4.4 Gages. The contractor shall provide gages as necessary to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings.

4.5 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major and Minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:10Z
Check the source to verify that this is the current version before use.

4.6 Quality conformance inspection. Quality conformance inspection shall consist of verification of the characteristics classified on the drawings of DL 2519703, DL 2519704 or DL 2519705 and verification of the following requirements and tests:

Item	Nature of Requirement Verification	Applicable Paragraphs	Classification (WR-43)
Primary Component	Single lot	3.3, 4.3.3	(M101)
Primer	Age, loading and acceptance	3.4, 4.7.1	(C1)
Primer	X-ray	3.4.1, 4.7.2	(C2)
Delay Composition	Acceptance and loading	3.5	(C3)
Ignition Powder	Acceptance and loading	3.6	(C4)
Propellant	Acceptance and weight	3.7.1	(C5)
Smokeless Powder	Condition	3.7.3, 4.7.3	(C6)
Cartridge	Visual inspection	3.8.1, 4.7.4	(M102)
Cartridge	Leakage	3.8.2, 4.7.5	(C7)
Cartridge	X-ray	3.8.3, 4.7.6	(C8)
Cartridge	Serial number identification	3.8.3.1, 4.7.6	(M103)
Cartridge	Ballistic tests	3.9, 4.7.7	(C9)
Cartridge	Packaging, packing and container markings	3.10, 4.6.1	(M104)

4.6.1 Packaging, packing and marking. Quality conformance inspection shall be conducted in accordance with the inspection criteria on the applicable drawings referenced in Section 5.

4.7 Tests.

4.7.1 Primer verification. Verification shall be provided that the primers used were manufactured, loaded, and accepted in accordance with the requirements of 3.4.

4.7.2 Primer X-ray. Examine all primers by X-ray and reject those having any of the defects of 3.4.1. Pressure-sensitive tape may be used as an aid in positioning the primers in suitable grooved racks. For the most revealing exposure, the primers are laid on their side with the anvil cutouts vertical.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:10Z
Check the source to verify that this is the current version before use.

4.7.3 Smokeless powder, condition. Smokeless powder shall be inspected in accordance with the applicable paragraphs of OP 5. Evidence of decomposition shall be cause for rejection.

4.7.4 Visual inspection. Examine all cartridges and reject those having any of the defects of 3.8.1.

4.7.5 Leakage. Each cartridge in the preproduction sample or production lot sample shall be leak tested in a dry gas or dry air leak tester. Cartridges which exhibit a leak rate in excess of 10^{-5} cc/sec of gas or air at a pressure differential of 1.0 ± 0.1 atmosphere shall be considered defective. Cartridges of the preproduction sample which exhibit a leakage defect shall be rejected. If one or more cartridges of the production lot sample exhibit a leakage defect, then a 100 percent leak test of the lot being tested shall be required at no additional expense to the government. Cartridges which exhibit a leak rate in excess of that specified above shall be rejected.

4.7.6 X-ray examination. Examine all cartridges by X-ray and any observable imperfections in assembly shall be cause for cartridge rejection. The cartridges shall be positioned on their sides for the most revealing exposure. All cartridges shall be identified with serial numbers prior to X-ray examination. The cartridge serial numbers shall be in consecutive order beginning with the number one (1) in each production lot. The cartridges for X-ray shall be arranged on trays or boards in consecutive numerical order and each X-ray plate shall carry a permanent identification of the cartridges displayed thereon. The X-ray plate identification shall include the complete lot number, as stamped on the cartridges, and the span of serial numbers displayed. The serial numbers shall be rubber stamped on the cartridges with approximately 1/8 inch high characters using the same process as that applying to the other cartridge markings. The serial numbers shall be located apart from the other cartridge markings. X-ray plates of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. See 6.2.

4.7.7 Ballistic tests.

4.7.7.1 The number of cartridges selected from each production lot for the ballistic test shall be in accordance with the requirements of MIL-STD-105, Inspection Level II, for an AQL of 0.15, single sampling plan for normal inspection. The sample shall be randomly divided into four equal groups. If the sample does not divide evenly, the odd cartridges will be placed in the third group. Samples in Group 1 inserted in the sleeve (Drawing 2838190), of the test set (Drawing 2838183), shall be conditioned at -65° ± 5°F for a minimum of six hours and fired in the test set within two minutes after removal from conditioning. Samples

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:10Z
Check the source to verify that this is the current version before use.

in Group 2 inserted in the sleeve (Drawing 2838190), of the test set (Drawing 2838183), shall be conditioned at +160° ± 5°F for a minimum of six hours and fired in the test set within two minutes after removal from conditioning. Samples in Group 3 shall be subjected to the 28-day temperature and humidity test of MIL-D-21625 and after conditioning for a minimum of six hours at 70° ± 5°F shall be fired in the test set, Drawing 2838183. Samples in Group 4 shall be subjected to the vibration test of MIL-D-21625 and after conditioning for a minimum of six hours at 70° ± 5°F shall be fired in the test set, Drawing 2838183. In addition to the sample of cartridges for the above tests, ten additional cartridges shall be furnished from the production lot for investigative purposes.

4.7.7.2 Criteria of acceptability. Cartridges of the ballistic sample shall meet the requirements of 3.9. Acceptance for the delay time requirement of 3.9.1, the minimum pressure requirement of 3.9.2, and the time to peak pressure requirement of 3.9.3 shall be in accordance with MIL-STD-105 using an AQL of 0.15 applied separately to the requirement of each paragraph.

4.7.7.2.1 Misfire. The failure of any cartridge to fire shall result in rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.7.7.3 Retest. There shall be no retests.

4.7.7.3.1 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridges, original test results should be discarded and that part of the test reconducted.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. For definitions of packaging levels, see 6.4.

5.1.1 Level A. Level A packaging shall be used for packaging of all production lots for service use. Four cartridges shall be packaged in an inner container in accordance with the drawings of LD 537902.

5.1.2 Level B. Not applicable.

5.1.3 Level C. When directed by the procuring activity, those cartridges which are intended for expenditure in preproduction sample testing and for production lot acceptance tests shall be packaged in accordance with Method III of MIL-P-116 and with the Code of Federal Regulations 49 CFR 171-178.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:10Z
Check the source to verify that this is the current version before use.

5.2 Packing. For definitions of packing levels, see 6.4.

5.2.1 Level A. Level A packing shall be used for packing of all production lots for service use. Six inner containers as described in 5.1.1 shall be packed in accordance with the drawings of LD 537903.

5.2.2 Level B. Not applicable.

5.2.3 Level C. When directed by the procuring activity, cartridges intended for expenditure in preproduction sample testing and for production lot acceptance tests, and are packaged in accordance with 5.1.3, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with the Code of Federal Regulations 49 CFR 171-178.

5.3 Markings.

5.3.1 Special markings. Marking of exterior containers shall be in accordance with the Code of Federal Regulations 49 CFR 171-178.

5.3.2 Normal markings. Unless otherwise specified in the contract or order, the marking information on unit packages and shipping containers shall be as specified on the drawings of LD 537902 and LD 537903, respectively. The specified marking information shall be applied to the containers in accordance with the applicable provisions on MIL-STD-129.

6. NOTES

6.1 Intended use. The MARK 4 MOD 2, MARK 5 MOD 2 and MARK 6 MOD 2 delay cartridges were originally designed for use in automatic lap-belt and parachute opening actuators in conjunction with emergency escape from aircraft.

6.2 Ordering information. Procurement documents should specify the following:

a. title, number and date of this specification;

b. whether preproduction inspection is required and, if so, specify the test activity;

c. production lot size and test activity (see 4.3.3);

d. direction for shipping X-ray plates of entire production lot along with ballistic sample (paragraph 4.7.7.1) to the activity conducting production lot acceptance test;

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:10Z
Check the source to verify that this is the current version before use.

e. items of data required for each preproduction and production lot (see 6.3).

6.3 Data. For the information of contractors and contracting officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in Section 2 of this specification or (c) referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order.

6.3.1 Test report. Results of ballistic tests shall be forwarded to the Naval Air Systems Command (AIR-53222) and the procuring activity.

6.3.2 Data cards. The contract or order should state the format, quantity and destination of data cards.

6.4 Definitions. For the purposes of this specification, the following definitions apply.

6.4.1 Level A packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during handling, shipment, indeterminate storage and world-wide redistribution.

6.4.1.1 Level C packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to supplier's commercial practice when such meets the requirements of this level.

6.4.2 Level A packing. The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and world-wide redistribution.

6.4.2.1 Level C packing. The degree of packing which will afford protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

6.5 Explosives safety precautions. Safety precautions for explosive-loaded items are contained in MIL-STD-1314.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:10Z
Check the source to verify that this is the current version before use.

6.5.1 Applicability. The loading, assembly and handling of the explosives, explosive-loaded subassemblies and the finished items covered by this specification involve hazardous operations and therefore require suitable explosives safety precautions. At plants of the Naval Air Systems Command, Department of the Navy, and other Government plants where the Government is responsible for safety, the precautions listed in MIL-STD-1314 are mandatory. These same minimum provisions are recommended for observance by a contractor or manufacturer that explosively loads or assembles the item covered in this specification in order to fulfill his own safety responsibility.

Project No. 1377-N403

☆ U. S. GOVERNMENT PRINTING OFFICE: 1970 -433-690/1668

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:10Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-C-81788(AS) CARTRIDGES, DELAY, MARK 4 MOD 2 MARK 5 MOD 2 AND MK 6 MOD 2

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426

C-8279

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:10Z
Check the source to verify that this is the current version before use.